



M²RNet: Multi-modal and multi-scale refined network for RGB-D salient object detection

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ARTICLE INFO

Article history:

Received 22 July 2022

Revised 11 October 2022

Accepted 2 November 2022

Available online 5 November 2022

Keywords:

Saliency detection

Deep learning

Multi-modal feature

Multi-scale feature

Loss function

ABSTRACT

Salient object detection is a fundamental topic in computer vision, which has promising application prospects. The previous methods based on RGB-D may potentially suffer from the incompatibility of multi-modal feature fusion and the insufficiency of multi-scale feature aggregation. To tackle these two dilemmas, we propose a novel multi-modal and multi-scale refined network (M²RNet). Specifically, three essential components are presented in this network. The nested dual attention module (NDAM) explicitly exploits the combined features of RGB and depth flows. The adjacent interactive aggregation module (AIAM) gradually integrates the neighbor features of high, middle and low levels. The joint hybrid optimization loss (JHOL) makes the predictions have a prominent outline. Extensive experiments quantitatively and qualitatively demonstrate that our method outperforms other state-of-the-art approaches.

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1. Introduction

Salient object detection (SOD) aims to identify the most conspicuous object that attracts humans in the scene, which not only feeds concise and useful information for subsequent computer vision tasks, but also saves the computational overhead. It has been successfully applied in various fields, such as image retrieval [1,2], robot navigation [3], person re-identification [4] and many more.

The existing SOD methods have exhibited abroad prospects of application benefiting from the powerful representation ability of convolutional neural networks (CNNs) [5], especially fully convolutional networks (FCNs) [6]. Most of them resort to a single RGB information, which is difficult to achieve satisfactory results in complex scenes. At present, depth information has become growing popular thanks to the emergence of affordable and portable devices. As a supplement to RGB features, depth features provide rich distance information. However, the inherent differences between the multiple modalities lead to the bottleneck of feature fusion. Moreover, although the features of each scale contain the detail information or semantic information, it is hard to adequately aggregate them. As a result, the previous methods based on RGB-D may potentially suffer from the incompatibility of multi-modal feature fusion and the insufficiency of multi-scale feature aggregation.

To this end, we propose a novel multi-modal and multi-scale refined network (M²RNet) for RGB-D salient object detection. Specifically, the network is composed of three presented components of nested dual attention module (NDAM), adjacent interactive aggregation module (AIAM) and joint hybrid optimization loss (JHOL). In NDAM, we sequentially leverage the channel attention and spatial attention to explicitly understand what and where is meaningful, thereby emphasizing or suppressing RGB and depth information that is important or unnecessary, for the purpose of boosting the fusion of RGB and depth features. In AIAM, we leverage the interaction of the progressive connection and jumping connection in parallel to gradually learn information in abundant resolution for the purpose of boosting the aggregation of high level, middle level and low level features. In JHOL, we are devoted to guaranteeing inter-class discrimination and intra-class consistency by taking the local correlation and global correlation of each pixel into account. These three components work together to achieve remarkable object detection results. The visual comparison results of our method and two state-of-the-art approaches are shown in Fig. 1. Comparing with the saliency maps obtained by S²MA [7] and CPFP [8], those of our M²RNet are more exact, which are overwhelmingly approximate to ground truth.

Our main contributions are summarized as follows:

- We propose a multi-modal and multi-scale refined network (M²RNet), which is equipped with the presented nested dual attention module (NDAM), adjacent interactive aggregation

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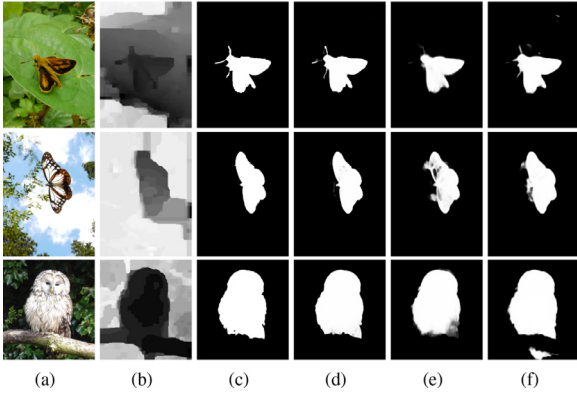


Fig. 1. Visual comparison of our method and two state-of-the-art approaches on several examples. (a) RGB; (b) Depth; (c) Ground truth; (d) Ours; (e) S²MA [7]; (f) CPFP [8].

module (AIAM) and joint hybrid optimization loss (JHOL) components. Our network is capable of refining the multi-modal and multi-scale features simultaneously, nearly without extra computing cost under specific supervision.

- We conduct extensive experiments on seven datasets and demonstrate that our method achieves consistently superior performance against 12 state-of-the-art approaches including three RGB salient object detection approaches and nine RGB-D salient object detection approaches in terms of six evaluation metrics.

2. Related work

2.1. RGB saliency detection

Lots of RGB salient object detection methods have been developed during the past decades.

For instance, Zhang et al. [10] proposed a PAGR, which selectively integrates multi-level contextual information. Liu et al. [11] proposed a PiCANet, which generates attention over the context regions for each pixel. Su et al. [12] proposed a BANet, which simultaneously enhances the feature selectivity at the boundaries and keeps the feature invariance at the interiors. Zhao et al. [13] proposed an EGNNet, which explores edge information to preserve salient object boundaries. Wu et al. [14] proposed a CPD, which uses cascaded partial decoder to discard low-level features. Liu et al. [15] proposed a PoolNet, which explores the potentials of pooling. Zhang et al. [16] proposed a CapSal, which uses image captioning for detecting. Qin et al. [17] proposed a BASNet, which focuses on end-to-end boundary-aware. Pang et al. [18] proposed a MINet, which exchanges information between multi-scale features. Wei et al. [19] proposed a LDF, which decouples saliency label into body map and detail map for iterative information exchange. Xu et al. [20] proposed a SA-DPNet, which learns the spatial-sensitive global context.

2.2. RGB-D saliency detection

Most recently, RGB-D salient object detection methods have rapidly aroused the concern of researchers and made impressive progress.

For instance, Chen et al. [21] proposed a TANet, which combines bottom-up stream and top-down stream to learn cross-modal complementarity. Wang and Gong [22] proposed an AFNet that adaptively fuses predictions from separate RGB and depth streams using switch mapping. Piao et al. [23] proposed a DMRA, which involves residual connection, multi-scale weighting and recurrent at-

tention. Zhao et al. [8] proposed a CPFP by making use of feature fusion of contrast prior and fluid pyramid. Fan et al. [24] proposed a D³Net, which automatically discards low-quality depth maps via gate connection. Li et al. [25] proposed an ICNet, which can learn the optimal conversion of RGB features and depth features to autonomously merge them. Pang et al. [26] proposed a HDFNet, in which the features of the network are densely connected and through the dynamic expansion pyramid. Fan et al. [27] proposed a BBS-Net, in which multi-level features are partitioned into teacher and student features in the cascade network. Zhao et al. [28] proposed a DANet, which explores early fusion and middle fusion between RGB and depth. Zhou et al. [29] proposed a SP-Net, which explores both the shared information and modality-specific properties. Li et al. [30] proposed a CMWNet, which weights the fusion of low, medium and high levels to encourage feature interaction. Fu et al. [31] proposed a JL-DCF for joint learning and densely-cooperative fusion. Zhang et al. [32] proposed an UC-Net, which learns the distribution of saliency maps by conditional variational autoencoder. Piao et al. [33] proposed an A2dele, which uses network prediction and attention as two bridges to transfer deep knowledge from deep stream to RGB stream. Liu et al. [7] proposed a S²MA, which reweights mutual attention for filtering out unreliable modality information. Zhang et al. [34] proposed a DFM-Net, which benefits from explicitly control and enhance depth features during cross-modal fusion. Li et al. [35] proposed a HAINet, which uses the hierarchical manner from RGB to depth and then back to RGB, so that local and global contexts are progressively integrated within a single feature scale. Wen et al. [36] proposed a DSNet, which automatically explores the cross-modal complementary information, bi-directionally optimizes the cross-level information, and adaptively strengthens the multi-scale information. For a more complete overview of RGB-D salient object detection, please refer to [37–39].

3. Proposed method

3.1. Network overview

The proposed multi-modal and multi-scale refined network (M²RNet) is an encoder-decoder architecture, covering nested dual attention module (NDAM), adjacent interactive aggregation module (AIAM) and joint hybrid optimization loss (JHOL), as shown in Fig. 2. To be concise, we formally denote the output features of RGB branch in the encoder as f_d^i ($i = 1, 2, 3, 4, 5$), the output features of depth branch in the encoder as f_{rgb}^i ($i = 1, 2, 3, 4, 5$), and the output features in the decoder as f_{rgbd}^i ($i = 1, 2, 3, 4, 5$). Let f_{cm}^i ($i = 3, 4, 5$) denote the combined features of RGB flow and depth flow, we utilize NDAM to strengthen their robustness to get the corresponding enhanced features f_{cm}^i ($i = 3, 4, 5$). For each group of three consecutive features $\{f_{rgb}^{i-1}, f_{rgb}^i, f_{rgb}^{i+1}\}$ ($i = 2, 3, 4$), we utilize AIAM to produce their aggregated features $f_{rgb'}^i$ ($i = 2, 3, 4$). In order to facilitate the optimization, we embed JHOL as the auxiliary loss. Figure 3 displays the feature fusion process of the decoder.

3.2. Nested dual attention module

There are some main issues in fusing RGB and depth features. The key point is that RGB and depth features are incompatible to a certain extent, which is due to the inherent differences between the two modalities. Besides, low-quality depth maps also inevitably bring more noise than cues.

In view of these, we propose a nested dual attention module (NDAM) to promote the coordination of multi-modal features.

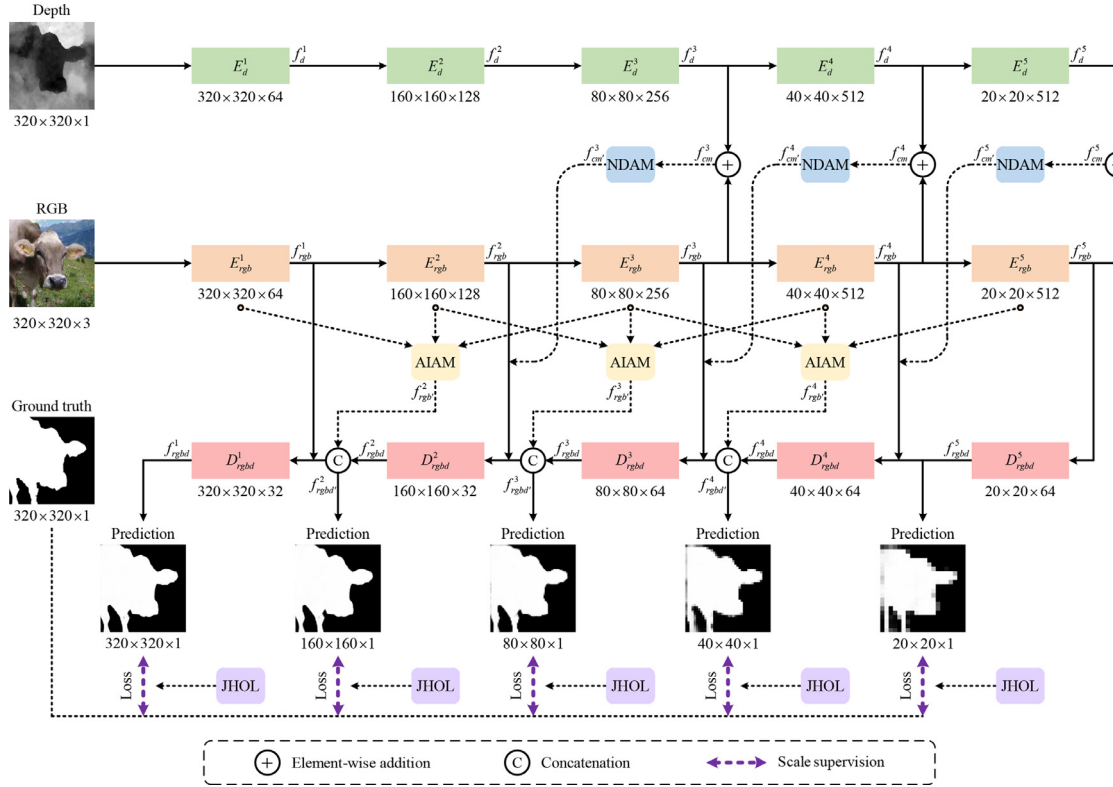


Fig. 2. The overall architecture of the proposed M²RNet. The backbone of our network is VGG-16 [9]. NDAM is used to explicitly exploit the combined features of RGB and depth flows. AIAM is used to gradually integrate the neighbor features of high, middle and low levels. JHOL is used to make the predictions have a prominent outline.

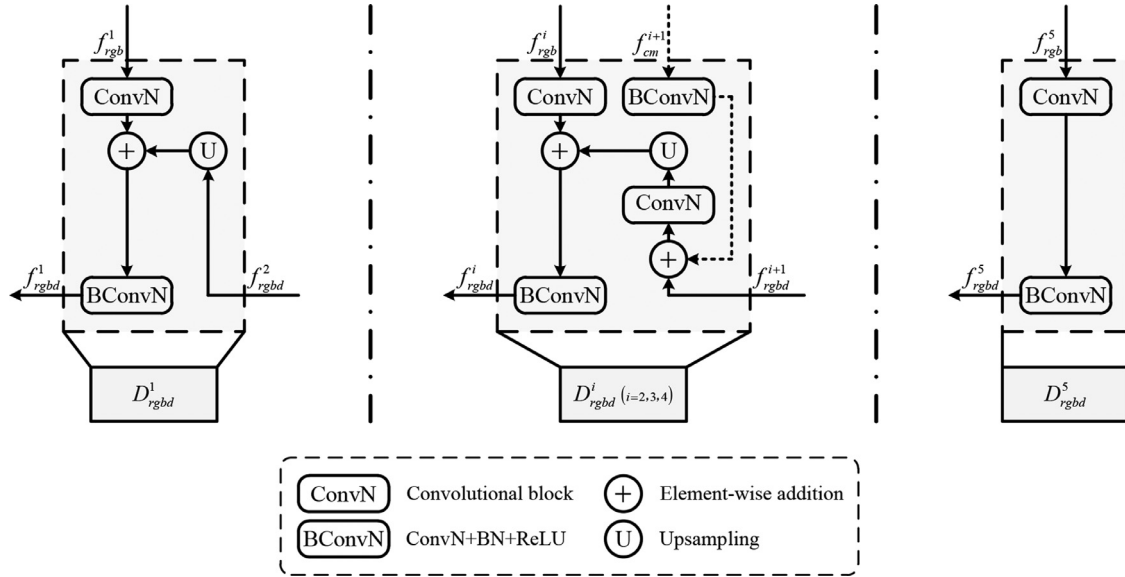


Fig. 3. Details of feature fusion in decoder.

As an incidental benefit, it can also reduce the noise contamination of depth map. The two-phase attention mechanism is elaborately designed to mine potential features, and the dual attention is employed in them at the same time. The outstanding advantage of dual attention is that it can improve the features not only from the channel relationship, but also from the spatial relationship. Here, the channel attention mechanism of each phase is responsible for excavating the channel relationship between features, while the spatial attention mechanism of each phase is responsible for excavating the spatial relationship between features. By

directly merging the RGB features f_{rgb}^i ($i = 3, 4, 5$) and the depth features f_d^i ($i = 3, 4, 5$), the combined features f_{cm}^i ($i = 3, 4, 5$) are easily calculated. Furthermore, the corresponding enhanced features $f_{cm'}^i$ ($i = 3, 4, 5$) are eventually obtained after the reinforcement of nested attention. The procedure of RGB and depth feature fusion can be described as:

$$f_{cm}^i = f_{rgb}^i + f_d^i, \quad (1)$$

$$f_{cm'}^i = S_2(C_2(S_1(C_1(f_{cm}^i))))), \quad (2)$$

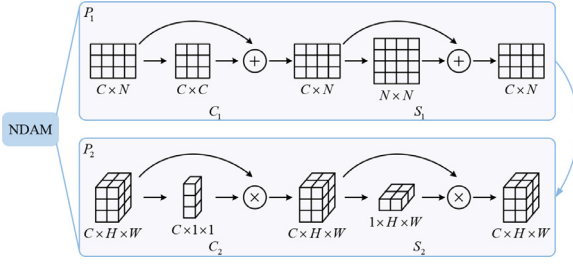


Fig. 4. Illustration of NDAM. P_1 and P_2 are its two phases, which are in order.

where $C_i(\cdot)$ and $S_i(\cdot)$ ($i = 1, 2$) denote the channel attention and spatial attention, respectively. The nested attention is roughly divided into two phases of P_1 and P_2 , as shown in Fig. 4.

In the first phase P_1 , given an intermediate feature $\bar{f} \in \mathbb{R}^{C \times N}$, in which $N = H \times W$, and C , H and W are the channel, height and width of the feature $\bar{f}$, respectively, the dual attentions $C_1(\cdot)$ and $S_1(\cdot)$ are defined as:

$$C_1(\bar{f}) = \delta(\text{Conv}(\bar{f}) \times \text{Conv}(\bar{f})^T) \times \text{Conv}(\bar{f}) + \bar{f}, \quad (3)$$

$$S_1(\bar{f}) = \text{Conv}(\bar{f}) \times \delta(\text{Conv}(\bar{f})^T \times \text{Conv}(\bar{f})) + \bar{f}, \quad (4)$$

where $(\cdot)^T$ denotes the transpose operation and $\delta(\cdot)$ is the Softmax function. Note that the channel of the feature in Eq. (4) is set to $1/8$ of the original channel of that for computation efficiency.

In the second phase P_2 , for the feature $\tilde{f} \in \mathbb{R}^{C \times H \times W}$, which can be reshaped with the feature $\bar{f}$, the dual attentions $C_2(\cdot)$ and $S_2(\cdot)$ are defined as:

$$C_2(\tilde{f}) = \sigma(\text{MLP}(\text{GMP}(\tilde{f}))) \odot \tilde{f}, \quad (5)$$

$$S_2(\tilde{f}) = \sigma(\text{Conv}(\text{GMPC}(\tilde{f}))) \odot \tilde{f}, \quad (6)$$

where $\odot$ denotes the element-wise multiplication, $\sigma(\cdot)$ is the Sigmoid function, $\text{MLP}(\cdot)$ represents the multi-layer perceptron, $\text{GMP}(\cdot)$ and $\text{GMPC}(\cdot)$ represent the global max pooling operation and global max pooling along the channel operation, respectively. Note that we use a global max pooling rather than a global average pooling since our goal is to find the area with the biggest visual influence.

3.3. Adjacent interactive aggregation module

In general, shallower features have more detail information, while deeper features have more semantic information. The aggregation of multi-level features with different resolutions enables the context information to be integrated as sufficient as possible.

Relying on this, we propose an adjacent interactive aggregation module (AIAM) to guide the interaction of multi-scale features. In this way, the neighbor features that are of intimate correlation are constantly complemented.

Assuming that h , m and l represent the high layer, middle layer and low layer, respectively, the relationship between the features from them can be fixed in triples. The interaction behind the features of three triples $\{f_{rgb}^{i-1}, f_{rgb}^i, f_{rgb}^{i+1}\}$ ($i = 2, 3, 4$) helps the generation of desired resulting features $\{f_{rgb}^i\}$ ($i = 2, 3, 4$). The procedure of different levels feature aggregation can be described as:

$$f_{rgb'}^i = I_1(f_{rgb}^{i-1}, f_{rgb}^i, f_{rgb}^{i+1}) + I_2(f_{rgb}^{i-1}, f_{rgb}^i, f_{rgb}^{i+1}), \quad (7)$$

$$f_{rgbd'}^i = \text{Conv}(\text{Cat}(f_{rgbd}^i, f_{rgb'}^i)), \quad (8)$$

where I_1 and I_2 denote two kinds of feature interactions, as shown in Fig. 5. The difference between them is that the interaction of

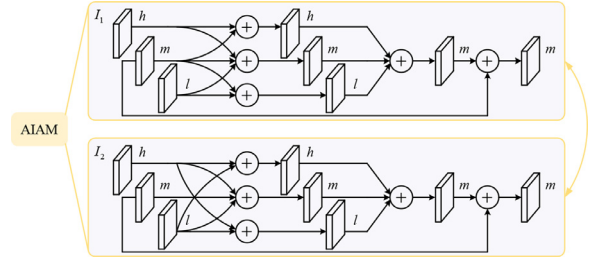


Fig. 5. Illustration of AIAM. I_1 and I_2 are its two kinds of interactions, which have no order.

the former is progressive, while that of the latter is jumping in the initial stage. After that, these relevant features repeatedly pass through some convolutional layers, batch normalization layers and ReLU layers, and finally are associated with a residual block.

3.4. Joint hybrid optimization loss

Let $\mathcal{P} = \{p | 0 < p < 1\}$ and $\mathcal{G} = \{g | 0 < g < 1\}$ denote the prediction saliency map and ground truth saliency map, respectively. As the most classical loss function, binary cross entropy loss (BCEl), symbolized by $\mathcal{L}_{BCE}$, can be formulated as:

$$\mathcal{L}_{BCE} = - \sum_{h=1}^H \sum_{w=1}^W (g \log p + (1-g) \log(1-p)), \quad (9)$$

where H and W are the height and width of the image, respectively.

To reinforce the capability of supervision from both local and global aspects, we propose a joint hybrid optimization loss (JHOL), symbolized by $\mathcal{L}_{JHO}$, as an auxiliary loss function, which can be formulated as:

$$\mathcal{L}_{JHO} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2 + \lambda_3 \mathcal{L}_3 + \lambda_4 \mathcal{L}_4, \quad (10)$$

where λ_1 , λ_2 , λ_3 and λ_4 are the parameters to control the trade-off among the four terms of the loss. For the sake of simplicity, they are all set to 1. To be specific, $\mathcal{L}_1$, $\mathcal{L}_2$, $\mathcal{L}_3$ and $\mathcal{L}_4$ are defined as:

$$\mathcal{L}_1 = \frac{\sum_{h=1}^H \sum_{w=1}^W (p(1-g))}{\sum_{h=1}^H \sum_{w=1}^W p}, \quad (11)$$

$$\mathcal{L}_2 = \frac{\sum_{h=1}^H \sum_{w=1}^W (g(1-p))}{\sum_{h=1}^H \sum_{w=1}^W g}, \quad (12)$$

$$\mathcal{L}_3 = \frac{\sum_{h=1}^H \sum_{w=1}^W (p(1-g) + g(1-p))}{\sum_{h=1}^H \sum_{w=1}^W (p + g - pg)}, \quad (13)$$

$$\mathcal{L}_4 = \frac{\sum_{h=1}^H \sum_{w=1}^W ((1-p)(1-g))}{\sum_{h=1}^H \sum_{w=1}^W (1 - p(1-g) - g(1-p))}, \quad (14)$$

where $\mathcal{L}_1$ and $\mathcal{L}_2$ along with $\mathcal{L}_3$ and $\mathcal{L}_4$ separately optimize errors from different levels in the form of two groups. The first two impose constraints on the errors of a single foreground or background class, and the latter two impose constraints on errors of a combined foreground and background classes. On the

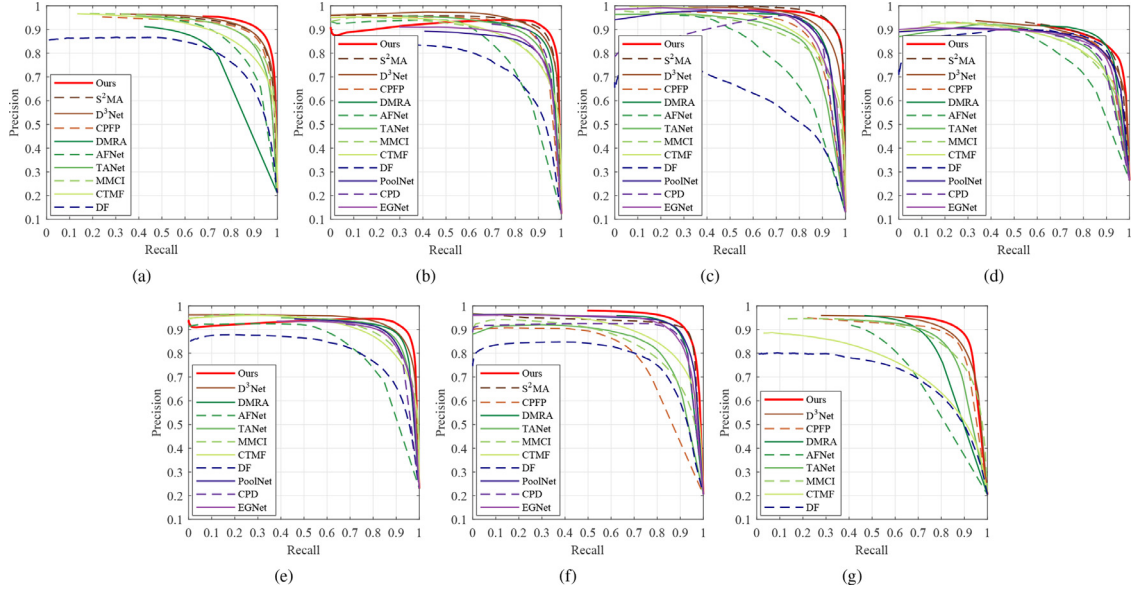


Fig. 6. The PR curves of different methods on seven datasets. (a) STEREO [43]; (b) NLPR [44]; (c) RGBD135 [45]; (d) LFSD [46]; (e) NJU2K [47]; (f) DUT-RGBD [23]; (g) SIP [24].

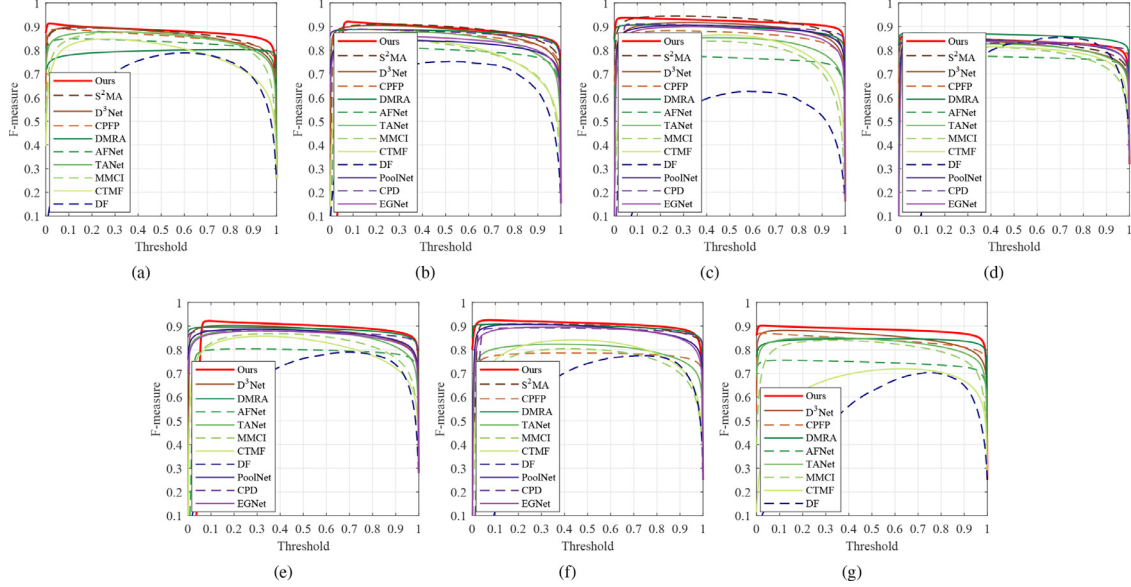


Fig. 7. The F_β curves of different methods on seven datasets. (a) STEREO [43]; (b) NLPR [44]; (c) RGBD135 [45]; (d) LFSD [46]; (e) NJU2K [47]; (f) DUT-RGBD [23]; (g) SIP [24].

one hand, $\mathcal{L}_1$ and $\mathcal{L}_2$ focus on the local correlation of each pixel to ensure a certain degree of discrimination. On the other hand, $\mathcal{L}_3$ and $\mathcal{L}_4$ focus on the global correlation of each pixel to ensure a certain degree of consistency. As a result, the items in $\mathcal{L}_{JHO}$ jointly makes the model capture the foreground region as perfectly as possible and filter the background region as steadily as possible.

Since BCEL is a pixel-level loss, the error is managed from the perspective of position. In contrast, JHOL manages error constraints locally and globally from the perspective of region. These two types of losses are integrated and optimized together to enhance the strength of supervision. Therefore, the total loss $\mathcal{L}_{Total}$ can be written as:

$$\mathcal{L}_{Total} = \mathcal{L}_{BCE} + \mu \mathcal{L}_{JHO}, \quad (15)$$

where μ is the parameter to control the trade-off between $\mathcal{L}_{BCE}$ and $\mathcal{L}_{JHO}$. In practice, it is also set to 1.

4. Experiments

4.1. Experimental setup

Datasets. We choose seven benchmark datasets including STEREO [43], NLPR [44], RGBD135 [45], LFSD [46], NJU2K [47], DUT-RGBD [23] and SIP [24] as the experimental material. STEREO, also known as SSB1000, contains 1000 pairs of stereoscopic images gathered from the Internet. NLPR contains 1000 images taken under different illumination conditions. RGBD135 is also called DES, which contains 135 images about some indoor scenarios. LFSD is relatively small and contains 100 images. NJU2K contains 1985 images collected from the Internet, 3D movies and photographs. DUT-RGBD contains 1200 images taken in varied real-life situations. SIP is relatively new, which contains 929 human images. Following [26], we use 700 samples from NLPR, 1485 samples from NJU2K,

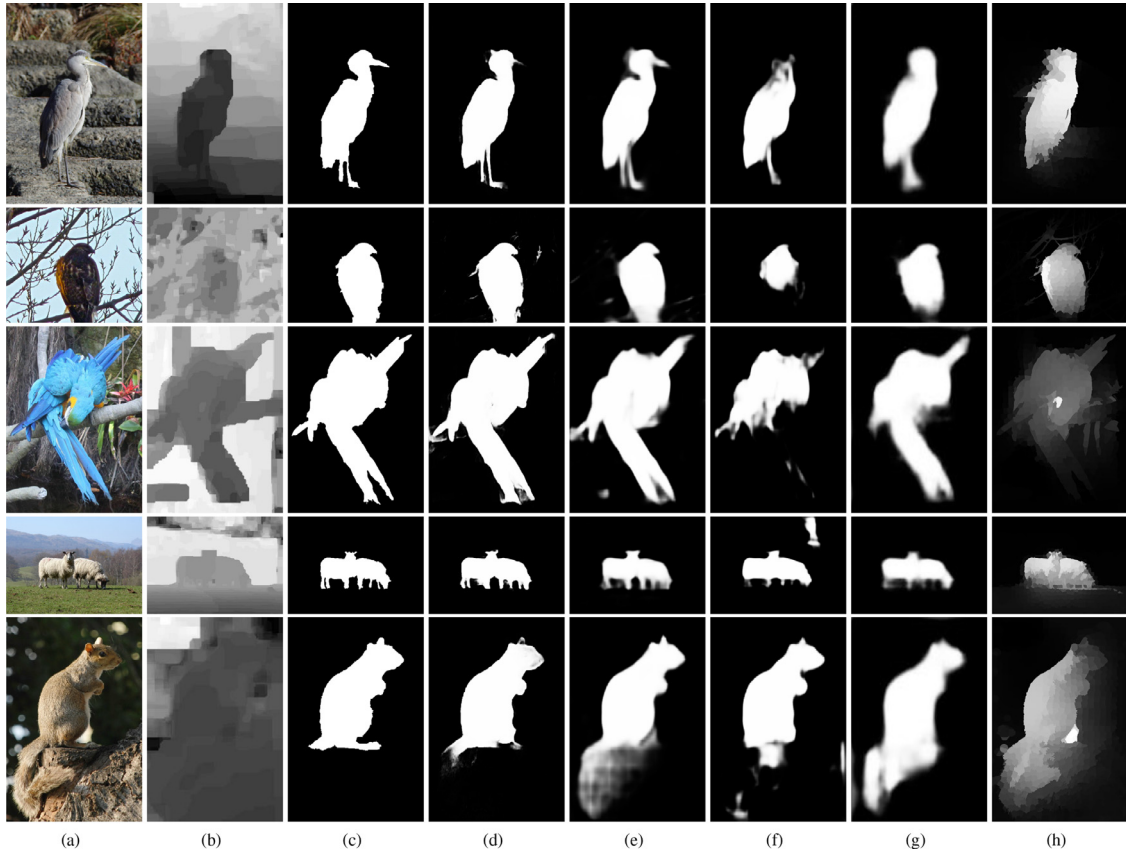


Fig. 8. Qualitative comparison of some methods on several representative examples. (a) RGB; (b) Depth; (c) Ground truth; (d) Ours; (e) D³Net [24]; (f) DMRA [23]; (g) TANet [21]; (h) DF [40].

and 800 samples from DUT-RGBD as the training set. The remaining samples and other datasets are used as the testing set.

Evaluation Metrics. We employ six typical evaluation metrics including S-measure (S_α) [49], maximum F-measure ($F_\beta^{\max}$) [50], average F-measure (F_β^{avg}) [50], weighted F-measure (F_β^w) [51], E-measure (E_ξ) [52] and mean absolute error ($\mathcal{M}$) [53] to comprehensively evaluate the performance of competitors. In addition, we also plot the precision-recall (PR) curves and F-measure (F_β) curves.

Implementation Details. We implement our method based on the PyTorch toolbox with a single GeForce RTX 2080 Ti GPU. The VGG-16 [9] is adopted as the backbone. For each input image, it is simply resized to 320×320 and then fed into the network to obtain prediction without any other pre-processing (e.g., HHA [54]) or post-processing (e.g., CRF [55]). To avoid over-fitting, the techniques of flipping, cropping and rotation act as data augmentation. The stochastic gradient descent (SGD) optimizer is used with the batch size of 4, the momentum of 0.9 and the weight decay of 5e-4. The whole network is stopped after 30 epochs.

4.2. Comparison with state-of-the-arts

The proposed method is compared with other 12 state-of-the-art approaches including three RGB approaches (i.e., EGNNet [13], CPD [14] and PoolNet [15]) and nine RGB-D approaches (i.e., DF [40], CTMF [41], MMCI [42], TANet [21], AFNet [22], DMRA [23], CPPF [8], D³Net [24] and S²MA [7]). For fair comparisons, all saliency maps of these methods are provided by the authors or computed by their released codes with default settings.

Quantitative Comparison. The quantitative comparison results of different methods on seven datasets in terms of six evalua-

tion metrics are reported in Table 1. It can be seen that our method performs best in almost all cases. Specifically, the proposed method achieves the best results on other datasets except LFSD, which can be attributed to the fact that this dataset is the smallest in all datasets. For an intuitive comparison, the PR curves and F_β curves of different methods on seven datasets are shown in Figs. 6 and 7, respectively. Obviously, the curves generated by our method are closer to the top and straighter in a large range than others, which reflects its excellence and stability.

Qualitative Comparison. The qualitative comparison results of some methods on several representative examples as shown in Fig. 8. It can be observed that our method can handle a wide variety of challenging scenes, such as blurred foreground, cluttered background, low contrast and multiple objects. More specifically, our method yields clear foreground, clean background, complete structure and sharp boundary. These results prove that our method is able to utilize cross-modal complementary information, which can not only achieve the reinforcement from the reliable depth maps but also prevent the contamination from the unreliable depth maps.

Comparison of Speed and Size. The results of speed and size comparison of advanced methods are reported in Table 2. It can be seen that our method reaches a real-time speed of 58 FPS, ranking first among these methods. At the same time, the model size of the proposed method is 172 MB, which is also ahead of other methods.

4.3. Ablation studies

A series of ablation studies are conducted to investigate the impact of each core component of our method. The ablation study results of our method in terms of uniform evaluation metrics are

Table 1

Quantitative comparison of different methods on seven datasets in terms of six evaluation metrics. $\uparrow$ and $\downarrow$ indicate that the larger and smaller scores are better, respectively. The best three results are highlight in red, blue and green. $\dagger$ and $\ddagger$ mean using VGG-19 [9] and ResNet-50 [48] as the backbone, respectively. If not marked, VGG-16 [9] is used as the backbone. '-' means no data available.

Dataset	Evaluation metric	EGNet [13]	CPD [14]	PoolNet [15]	DF [40]	CTMF [41]	MMCI [42]	TANet [21]	AFNet [22]	DMRA † [23]	CPFP [8]	D 3 Net ‡ [24]	S 2 MA [7]	M 2 RNet (ours)
STEREO [43]	$S_\alpha \uparrow$	-	-	-	0.757	0.848	0.873	0.871	0.825	0.752	0.879	0.891	0.890	0.899
	$F_\beta^{\max} \uparrow$	-	-	-	0.789	0.848	0.877	0.878	0.848	0.802	0.889	0.897	0.895	0.913
	$F_\beta^{\text{avg}} \uparrow$	-	-	-	0.742	0.771	0.829	0.835	0.807	0.762	0.830	0.833	0.855	0.867
	$F_\beta^\omega \uparrow$	-	-	-	0.549	0.698	0.760	0.787	0.752	0.647	0.817	0.815	0.825	0.851
	$E_\xi \uparrow$	-	-	-	0.838	0.870	0.905	0.916	0.887	0.816	0.907	0.911	0.926	0.929
	$\mathcal{M} \downarrow$	-	-	-	0.141	0.086	0.068	0.060	0.075	0.086	0.051	0.054	0.051	0.042
NLPR [44]	$S_\alpha \uparrow$	0.867	0.885	0.867	0.769	0.860	0.856	0.886	0.798	0.899	0.888	0.905	0.915	0.918
	$F_\beta^{\max} \uparrow$	0.857	0.889	0.844	0.753	0.840	0.841	0.876	0.816	0.888	0.888	0.905	0.910	0.921
	$F_\beta^{\text{avg}} \uparrow$	0.800	0.840	0.791	0.682	0.723	0.729	0.795	0.746	0.855	0.821	0.832	0.846	0.862
	$F_\beta^\omega \uparrow$	0.774	0.829	0.771	0.524	0.691	0.688	0.789	0.699	0.846	0.819	0.833	0.855	0.848
	$E_\xi \uparrow$	0.910	0.925	0.900	0.838	0.869	0.871	0.916	0.884	0.942	0.923	0.932	0.937	0.941
	$\mathcal{M} \downarrow$	0.047	0.037	0.046	0.099	0.056	0.856	0.041	0.060	0.031	0.036	0.034	0.030	0.033
RGBD135 [45]	$S_\alpha \uparrow$	0.876	0.891	0.886	0.681	0.863	0.848	0.858	0.770	0.900	0.872	0.904	0.941	0.934
	$F_\beta^{\max} \uparrow$	0.900	0.910	0.906	0.626	0.865	0.839	0.853	0.775	0.907	0.882	0.917	0.944	0.937
	$F_\beta^{\text{avg}} \uparrow$	0.843	0.869	0.864	0.573	0.778	0.762	0.795	0.730	0.866	0.829	0.876	0.906	0.910
	$F_\beta^\omega \uparrow$	0.780	0.824	0.807	0.383	0.686	0.650	0.739	0.641	0.843	0.787	0.831	0.892	0.903
	$E_\xi \downarrow$	0.930	0.930	0.940	0.806	0.911	0.904	0.919	0.874	0.944	0.927	0.956	0.974	0.971
	$\mathcal{M} \downarrow$	0.037	0.032	0.032	0.132	0.055	0.065	0.046	0.068	0.030	0.038	0.030	0.021	0.019
LFSD [46]	$S_\alpha \uparrow$	0.818	0.806	0.826	0.776	0.796	0.787	0.801	0.738	0.847	0.828	0.832	0.837	0.842
	$F_\beta^{\max} \uparrow$	0.838	0.834	0.846	0.854	0.815	0.813	0.827	0.780	0.872	0.850	0.849	0.862	0.861
	$F_\beta^{\text{avg}} \uparrow$	0.803	0.808	0.790	0.811	0.780	0.779	0.786	0.742	0.849	0.813	0.801	0.820	0.825
	$F_\beta^\omega \uparrow$	0.745	0.753	0.757	0.618	0.695	0.663	0.718	0.671	0.811	0.775	0.756	0.772	0.786
	$E_\xi \uparrow$	0.854	0.856	0.852	0.841	0.851	0.840	0.845	0.810	0.899	0.867	0.860	0.876	0.874
	$\mathcal{M} \downarrow$	0.102	0.097	0.094	0.151	0.120	0.132	0.111	0.133	0.075	0.088	0.099	0.094	0.088
NJU2K [47]	$S_\alpha \uparrow$	0.869	0.862	0.872	0.735	0.849	0.859	0.878	0.771	0.886	-	0.895	-	0.910
	$F_\beta^{\max} \uparrow$	0.880	0.880	0.887	0.790	0.857	0.868	0.888	0.804	0.896	-	0.903	-	0.922
	$F_\beta^{\text{avg}} \uparrow$	0.846	0.853	0.850	0.744	0.779	0.803	0.844	0.766	0.872	-	0.819	-	0.841
	$F_\beta^\omega \uparrow$	0.808	0.821	0.816	0.553	0.731	0.749	0.812	0.699	0.853	-	0.839	-	0.854
	$E_\xi \downarrow$	0.905	0.908	0.908	0.818	0.864	0.878	0.909	0.846	0.921	-	0.891	-	0.904
	$\mathcal{M} \downarrow$	0.060	0.059	0.057	0.151	0.085	0.079	0.061	0.103	0.051	-	0.051	-	0.049
DUT-RGBD [23]	$S_\alpha \uparrow$	0.872	0.874	0.892	0.719	0.830	0.791	0.808	-	0.888	0.749	-	0.903	0.903
	$F_\beta^{\max} \uparrow$	0.897	0.892	0.907	0.775	0.842	0.804	0.823	-	0.908	0.787	-	0.909	0.925
	$F_\beta^{\text{avg}} \uparrow$	0.861	0.863	0.866	0.748	0.790	0.751	0.771	-	0.883	0.735	-	0.866	0.892
	$F_\beta^\omega \uparrow$	0.797	0.819	0.829	0.514	0.681	0.626	0.703	-	0.852	0.636	-	0.856	0.864
	$E_\xi \uparrow$	0.914	0.915	0.924	0.842	0.882	0.855	0.866	-	0.930	0.815	-	0.921	0.935
	$\mathcal{M} \downarrow$	0.060	0.059	0.050	0.150	0.097	0.113	0.093	-	0.048	0.100	-	0.046	0.042
SIP [24]	$S_\alpha \uparrow$	-	-	-	0.653	0.716	0.833	0.835	0.720	0.800	0.850	0.864	-	0.882
	$F_\beta^{\max} \uparrow$	-	-	-	0.704	0.720	0.840	0.851	0.756	0.847	0.870	0.882	-	0.902
	$F_\beta^{\text{avg}} \uparrow$	-	-	-	0.673	0.684	0.795	0.809	0.705	0.815	0.819	0.831	-	0.868
	$F_\beta^\omega \uparrow$	-	-	-	0.406	0.535	0.712	0.748	0.617	0.734	0.788	0.793	-	0.840
	$E_\xi \uparrow$	-	-	-	0.794	0.824	0.886	0.894	0.815	0.858	0.899	0.903	-	0.921
	$\mathcal{M} \downarrow$	-	-	-	0.185	0.139	0.086	0.075	0.118	0.088	0.064	0.063	-	0.049

Table 2

Comparison of speed and size of advanced methods.

Method	Speed (FPS)	Size (MB)
DMRA [23]	13	228
CPFP [8]	6	278
D 3 Net [24]	32	530
S 2 MA [7])	25	330
M 2 RNet (ours)	58	172

reported in Table 3. The baseline, corresponding to scheme No. 1 (i.e., the 1st row), refers to the network like FPNs [56]. It should be pointed out that the uniform evaluation metrics are redefined as a weighted sum of scores according to the proportion of each dataset in all datasets. The visual comparison results of the impact of each component of our method on several examples are shown in Fig. 9.

Effect of NDAM. Actually, NDAM consists of two parts, which are P_1 and P_2 . By adding them into the baseline in the individual and collective manner, corresponding to scheme No. 2-3 (i.e., the 2nd and 3rd rows) and scheme No. 4 (i.e., the 4th row), the performance is well improved. This confirms that NDAM does indeed

offer additional valuable information from the channel and spatial perspectives.

Effect of AIAM. Similarly, AIAM consists of two parts, which are I_1 and I_2 . By adding them into the baseline in the individual and collective manner, corresponding to scheme No. 5-6 (i.e., the 5th and 6th rows) and scheme No. 7 (i.e., the 7th row), the performance is greatly improved. This reveals the advantage of feature aggregation of AIAM.

Effect of JHOL. Also similarly, JHOL (i.e., $\mathcal{L}_{JHO}$) consists of four parts, which are $\mathcal{L}_1$, $\mathcal{L}_2$, $\mathcal{L}_3$ and $\mathcal{L}_4$. By adding them into the baseline in the individual and collective manner, corresponding to scheme No. 8-11 (i.e., the 8th, 9th, 10th and 11th rows) and scheme No. 12 (i.e., the 12th row), the performance is significantly improved. In particular, when these parts are collectively added into the baseline, there are 1.4%, 3.6%, 5.6%, 2.2%, 0.8% and 1.4% improvement in terms of uniform evaluation metrics in order, respectively. This suggests that the use of JHOL is crucial for the task.

4.4. Failure cases and analyses

Some failure cases of our method are shown in Fig. 10. There are two challenging scenes, including occlusion (i.e., the 1st row)

Table 3

Ablation study of our method in terms of uniform evaluation metrics. $\uparrow$ and $\downarrow$ indicate that the larger and smaller scores are better, respectively. The best results are **bold**.

No.	Baseline	+NDAM		+AIAM		+JHOL				$S_{\alpha} \uparrow$	$F_{\beta}^{\max} \uparrow$	$F_{\beta}^{\text{avg}} \uparrow$	$F_{\beta}^{\omega} \uparrow$	$E_{\xi} \uparrow$	$\mathcal{M} \downarrow$
		+ P_1	+ P_2	+ I_1	+ I_2	+ $\mathcal{L}_1$	+ $\mathcal{L}_2$	+ $\mathcal{L}_3$	+ $\mathcal{L}_4$						
1	✓									0.885	0.820	0.775	0.899	0.872	0.064
2	✓	✓								0.893	0.833	0.790	0.907	0.880	0.060
3	✓		✓							0.887	0.827	0.779	0.903	0.875	0.063
4	✓	✓	✓							0.894	0.837	0.796	0.910	0.882	0.059
5	✓			✓						0.896	0.839	0.804	0.914	0.885	0.058
6	✓				✓					0.890	0.832	0.788	0.906	0.878	0.061
7	✓			✓	✓					0.898	0.854	0.815	0.917	0.887	0.054
8	✓					✓				0.891	0.865	0.812	0.919	0.873	0.055
9	✓						✓			0.892	0.812	0.786	0.899	0.866	0.061
10	✓							✓		0.893	0.848	0.819	0.916	0.876	0.053
11	✓								✓	0.887	0.819	0.778	0.900	0.871	0.063
12	✓					✓	✓	✓	✓	0.899	0.856	0.831	0.921	0.880	0.050
13	✓	✓	✓	✓	✓	✓	✓	✓	✓	0.913	0.873	0.854	0.929	0.897	0.043

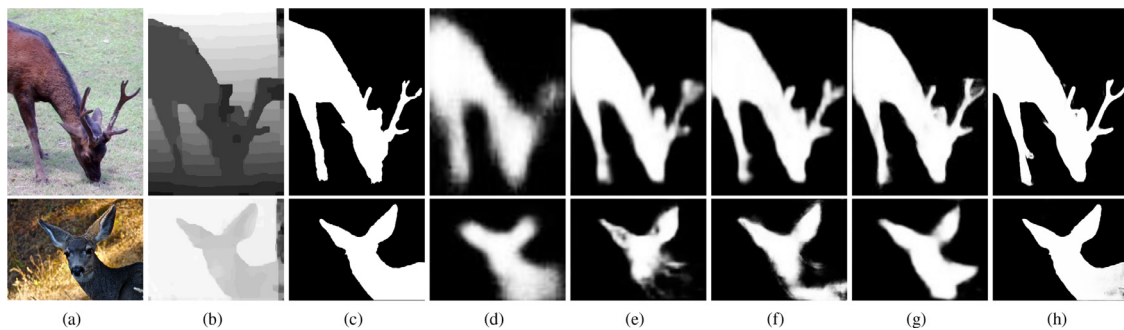


Fig. 9. Visual comparison of the impact of each component of our method on several examples. (a) RGB; (b) Depth; (c) Ground truth; (d) Baseline; (e) +NDAM; (f) +AIAM; (g) +JHOL; (h) +NDAM+AIAM+JHOL.

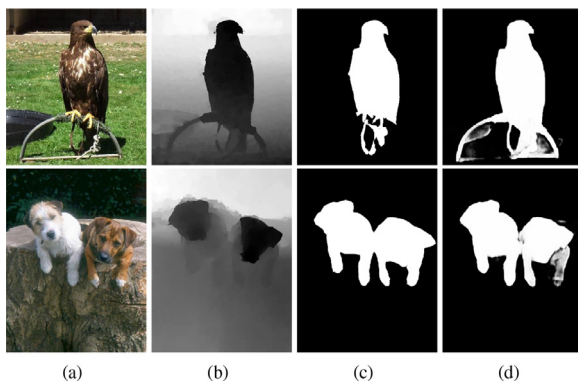


Fig. 10. Some failure cases of our method. (a) RGB; (b) Depth; (c) Ground truth; (d) Ours.

and multi-object (i.e., the 2nd row). For the first scene, the salient object is blocked by the distractor, while our method fails to accurately highlight the salient object, and incorrectly highlights the distractor. For the second scene, it contains more than two similar salient objects, but our method only highlights part of them. The reason for the failure is that these scenarios either tend to make the method unable to distinguish the hidden interference around the salient object, or tend to make the method suppress the salient object with long-distance information, which brings great difficulties to the saliency detection.

5. Conclusion

In this paper, we propose a novel multi-modal and multi-scale refined network, named M^2RNet , for detecting salient objects. To

start with, we present the nested dual attention module, which boosts the fusion of multi-modal features via the phased channel attention and spatial attention. Next, we present the adjacent interactive aggregation module, which boosts the aggregation of multi-scale features in the form of the progressive connection and jumping connection. Last but not least, we present the joint hybrid optimization loss, which alleviates the imbalance of pixels from both local and global aspects. Exhaustive experimental results demonstrate the superiority of our method over the other 12 state-of-the-art approaches. Although the proposed method can effectively alleviate the pollution caused by some low-quality depth maps, it cannot eliminate the problem from the root. In the future, we prepare to borrow the gating technology to selectively discard a small number of unqualified depth maps.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

Acknowledgments

This research was supported by the National Key R&D Program of China under Grant 2019YFB1311804, the [National Natural Science Foundation of China](#) under Grant [61973173](#), [91848108](#) and [91848203](#), the Technology Research and Development Program of Tianjin under Grant 18ZXNGX00340 and 20YFZCSY00830, and the

Tianjin Graduate Scientific Research Innovation Project under Grant 2021YJS02S02.

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